CSC 350 Assignment 2
Spring 2025

Due: Tuesday, February 4 by 11:59 pm

General Instructions

- Scan or compile snapshots of your submission into a PDF and upload it to Canvas.
- Assignments are **due before midnight** on the scheduled due date; generally, late assignments will not be accepted.
- Unless specified, **justify your answers always**. If justifications are missing or do not show reasonably how your answers are derived, 0 points will be assigned regardless of the correctness of the answers.
- Write **neatly** in the event you write your answers on paper. **Especially, for table completion questions, if your answers are not given neatly in well-drawn tables, the answers won't be graded**.
- **This assignment must be done by yourself without any unauthorized assistance**. Write your full name and numeric university ID on the front page of your submission, along with the following pledge statement.

I pledge my honor that all parts of this assignment were done by me individually and without any unauthorized assistance or collaboration with others.

Q1. Normal Form

Use equivalences to transform each of the following wff into a DNF. Write down the equivalence you use at each step. Next use truth tables to verify that the DNF formula you obtained is indeed equivalent to the wff that was given.

(a) $Q \wedge \neg P \rightarrow P$

(b) $(P \vee Q) \wedge R$

(c) $(A \vee B) \rightarrow (C \rightarrow D)$

(d) $((A \rightarrow B) \rightarrow C) \rightarrow \neg A$

Q2. Full DNF

Use equivalences to transform each of the following wff into a full DNF. Show full work to justify your answer.

(a) $P \rightarrow Q \wedge R$

(b) $(A \vee B) \wedge (A \vee C)$

Q3. Full CNF

Use equivalences to transform each of the following wff into a full CNF. Show full work to justify your answer. (Partial Credit) If you are unable to do the transformation using equivalences, then use truth table to obtain a full CNF.

(a) $P \rightarrow Q \wedge R$

(b) $(A \vee B) \wedge (A \vee C)$

(c) $(A \wedge \neg C) \vee \neg (B \rightarrow \neg C)$

Q4. Complete Sets

Consider a known set of connectives (such as $\{\neg, \vee\}$).

(a) Show that $\{\neg, \rightarrow\}$ is a complete set of connectives.

(b) Show that $\{NOR\}$ is a complete set of connectives.

HINT: For a given set of connectives (such as $\{NOR\}$), show that all connctives of the known set can be "implemented" using the given set.

Q5. Formal Proofs

For each of the following tautologies, provide a formal proof by using the CP rule.
Do **not** use the IP rule.

(a) $(A \vee B \rightarrow C) \wedge A \rightarrow C$

(b) $(A \rightarrow C) \rightarrow (A \rightarrow B \vee C)$

Now for the same tautologies, provide a formal proof by using the CP rule and by also using the IP rule at least once in each proof.

(a) $(A \vee B \rightarrow C) \wedge A \rightarrow C$

(b) $(A \rightarrow C) \rightarrow (A \rightarrow B \vee C)$